

Michael Chungyoun

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SUMMARY:

Fifth year PhD candidate at **Johns Hopkins University** in **Jeffrey Gray's lab** training **protein language models** and **curating therapeutic antibody data** from public resources for **therapeutic antibody discovery**. Invited speaker at **The AWS Life Science Symposium 2026**, **The Festival of Biologics 2024** and **European RosettaCon 2023**. Poster presenter at **NeurIPS MLSB 2023**. Published **16 scientific publications**, contributed to **two patents** in drug design and computational biology, and previously held **five internship/industry positions** at various biotechnology companies.

EDUCATION:

Johns Hopkins University, Baltimore, MD Sep 2021 – (Expected end) Dec 2026

- Ph.D. – **Chemical Engineering**, Funded by **NSF GRFP** and **GEM Fellowships**

University of Washington, Seattle, WA Sep 2016 – Jun 2021

- B.S. – **Chemical Engineering** | 5x Dean's List Award

RESEARCH:

Doctoral Researcher, Johns Hopkins University, Baltimore, MD, Prof. Jeffrey Gray Sep 2021 – Dec 2026

- Developing an antibody large language model capable of controllable sequence generation for therapeutic antibody discovery
- Curating publicly available antibody sequence and developability data for benchmarking and training developability prediction models

INDUSTRY:

Research Scientist, Sensible Biotechnologies, Boston, MA, Dr. Miroslav Gasparek Sep 2024 – Apr 2025

- Conducted prior art research on mRNA delivery and biomanufacturing technologies

Machine Learning Intern, Genentech/Prescient Design, New York, NY, Prof. Richard Bonneau May 2022 – Sep 2022

- Integrated proteinMPNN with distribution-free uncertainty quantification to screen antibody candidates for desirable therapeutic properties

Data Science Intern, Genentech, San Francisco, CA, Dr. Victor Saucedo Jun 2021 – Sep 2021

- Automated real-time quality assessment for large-molecule therapeutics using machine learning

Computational Biology Intern, Adaptive Biotechnologies, Seattle, WA, Dr. Paul Fields Nov 2020 – Jun 2021

- Performed next-generation sequencing and analyzed ssDNA nanostructures for drug delivery

Medical Diagnostics Intern, Novo Nordisk, Seattle, WA, Prof. Per Reinhall Oct 2020 – Jun 2021

- Developed 3-sensor (optical, electrical, and ultrasound) noninvasive blood pressure monitor

ENTREPRENEURSHIP:

AI Education Platform Founder, Hopkins AI for K-12 and Undergraduates, Baltimore, MD Jan 2021 – Jun 2022

- Developed AI education materials for high school summer programs in Baltimore City Public Schools

Innovation Fellow, Mt. Sinai Innovation Partners, New York, NY, Dr. Dov Shamir Sep 2021 – Dec 2021

- Evaluated biotech commercialization strategies, market viability, and patent landscape for investment opportunities

PUBLICATIONS:

1. Ritter S., Rand L., Karthick S., et al., **Chungyoun M.**, et al. "Decoding Bispecific Antibody Developability: Design Rules and Predictive Models from a 160-Member Library". [In review in *PNAS*]. (2026). ([link](#)).
2. Erasmus M. F., Bedinger D., Hopkins E., et al., **Chungyoun M.**, et al. "A blinded, prospective benchmark of in silico antibody discovery anchored to experimental affinity and developability". *Nature Biotechnology*. (2026). ([link](#)).
3. van Niekerk L., Moller J., Ritter S., et al., **Chungyoun M.**, et al. "2025 Ginkgo Datapoints Antibody Developability Competition Outcomes: Limited Model Performance and a Call for Data Standardization". *MAbs*. (2026). ([link](#)).

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4. Ai H., Arnold F., Barrick D., et al., **Chungyoun M.**, et al. "Protein engineering: Status report". [*In review in Protein Engineering, Design & Selection*]. (2026).
5. Chu L., Vogt J., **Chungyoun M.**, et al. "ProtAff: Protein Binding Affinity Prediction via LoRA-Finetuned ESM-2". *bioRxiv*. (2026). ([link](#)).
6. Ludwig L., **Chungyoun M.**, Gray J. "GermRL: Alleviating The Germline Bias In Autoregressive Antibody Language Models Through Reinforcement Learning". *bioRxiv*. (2026). ([link](#)).
7. **Chungyoun M.**, Gray J. "Fitness Landscape for Antibodies 2: Benchmarking Reveals That Protein AI Models Cannot Yet Consistently Predict Developability Properties". [*In review in Cell Patterns*]. (2025). ([link](#)).
8. **Chungyoun M.**, Au G., Carpentier B., Puvada S., Thomas C., Gray J. "DL4Proteins Jupyter Notebooks Teach how to use Artificial Intelligence for Biomolecular Structure Prediction and Design". *Biophysicist*. (2025). ([link](#)).
9. Escarra-Senmarti M., **Chungyoun M.**, Ferris D., Gray J., Andrade F. "Anti-citrullinated Protein Antibodies Arise During Affinity Maturation of Germline Antibodies to Carbamylated Proteins in Rheumatoid Arthritis". *bioRxiv*. (2025). ([link](#)).
10. Wang N., Tonko P., Ragav N., **Chungyoun M.**, et al. "A Perspective on K-12 AI Education". *Technology and Innovation*. (2024). ([link](#)).
11. Martins R., dos Santos A., da Costa C., Canner S., **Chungyoun M.**, et al. "Thermostability Enhancement of GH 62 α -l-Arabinofuranosidase by Directed Evolution and Rational Design". *Journal of Agricultural and Food Chemistry*. (2024). ([link](#)).
12. **Chungyoun M.**, Gray J. "AI Models for Protein Design are Driving Antibody Engineering". *Current Opinion in Biomedical Engineering*. (2023). ([link](#)).
13. **Chungyoun M.**, Ruffolo J., Gray J. "FLab: Benchmarking Tasks in Fitness Landscape Inference for Antibodies". *NeurIPS - Machine Learning in Structural Biology (MLSB) workshop*. (2023). ([link](#)).
14. McKenna M., Filteau J., Butler B., Sluis K., **Chungyoun M.**, et al. "Organotypic Whole Hemisphere Brain Slice Models to Study the Effects of Donor Age and Oxygen-Glucose-Deprivation on the Extracellular Properties of Cortical and Striatal Tissue". *Journal of Biological Engineering*. (2022). ([link](#)).
15. Liao R., Pon J., **Chungyoun M.**, Nance E. "Enzymatic Protection and Biocompatibility Screening of Enzyme-loaded Polymeric Nanoparticles for Neurotherapeutic Applications". *Biomaterials*. (2020). ([link](#)).
16. Janezic E., Lauer S., Williams R., **Chungyoun M.**, et al. "N-glycosylation of α 1D-adrenergic Receptor N-terminal Domain is Required for Correct Trafficking, Function, and Biogenesis". *Scientific Reports*. (2020). ([link](#)).

PATENTS:

1. Kelow S., Berenberg D., Nerli S., Leaver-Fay A., et al., **Chungyoun M.**, inventors; Genentech Global Patent Operations Team, assignee. "Hybrid Protein Design". *United States patent number WO2024011233A1*. (2024). ([link](#)).
2. **Chungyoun M.**, Shin A., Peng H., Shahukar S., inventors; CoMotion at the University of Washington, assignee. "Three-sensor Unobtrusive Blood Pressure Monitoring Device". *United States provisional patent serial number 63/193,509*. (2021). (expired).

NOTABLE GITHUB REPOSITORIES:

DL4Proteins Notebook Series (720 GitHub stars)

Dec 2024

- Jupyter notebooks that provide an introduction to the fundamental deep learning concepts and models currently utilized in the protein design space ([link](#))

Fitness Landscape for Antibodies (190 GitHub stars)

Jan 2024

- The largest public therapeutic antibody design benchmark encompassing 6 developability properties: Expression, thermostability, immunogenicity, aggregation, polyreactivity, and binding affinity ([link](#))